



Introduction to Cloudera Quick Start VM

What It Is...

It is a **virtual machine image** provided by the company [Cloudera](https://www.cloudera.com/). You can run it with **VMware**, **VirtualBox**, or **KVM**, and it comes with a complete Hadoop environment already installed.

What It Contains....

The VM comes with most of the major Hadoop ecosystem tools:

- Hadoop HDFS
- YARN
- MapReduce
- Hive
- HBase
- Pig
- Sqoop
- Flume
- Cloudera Manager (older version)

Everything is preconfigured so you can directly **start running Hadoop jobs and practice data engineering**.

Note that, Cloudera QuickStart VM is **now outdated** and is no longer officially maintained. It is still commonly used in universities for basic Hadoop teachings.

HOW TO START CLOUDERA

Step 1 – Start the Virtual Machine

Step 2 – Log in

When the VM loads:

- Username: cloudera
- Password: cloudera (if asked)

Step 3 – Start the Cloudera Services

- Open the terminal and run:

```
sudo /home/cloudera/cloudera-manager --express
```

- Wait 1-2 minutes for services to initialize.

Step 4 – Access Cloudera Manager (optional but recommended)

Open the browser inside the VM and go to: <http://localhost:7180>

Login:

- Username: admin
- Password: admin

You will now see: HDFS, YARN, Hive, Hue, Spark, Impala and other services.

Step 5 – Start Hadoop + Hive + Spark services

If some services show red/yellow, start them: Go to Cloudera Manager → Clusters

1. Click Start to start all services
2. Confirm "Start all services"

Activity

Task 1 – Upload Dataset to HDFS

Upload <dataset> into HDFS: **/user/student/student_performance/**

Code Hint

```
hdfs dfs -mkdir -p /user/cloudera/student/student_performance
hdfs dfs -put <dataset path> /user/cloudera/student/student_performance/
hdfs dfs -ls /user/student/student_performance/
```

Submit: Screenshot of file location.

Task 2 – Preview the Data in HDFS

Display the first 10 rows to verify the correct data format.

```
hdfs dfs -cat /user/cloudera/student/student_performance/Student_performance_data.csv | head -n 10
```

Submit: Screenshot of the head output.

Task 3 – Create Hive Database & External Table

Create a Hive database student_db and an external table pointing to HDFS.

Steps

1. Open Hive: `hive`
2. Create DB and use it:

```
CREATE DATABASE student_db;
```

```
USE student_db;
```

3. Create external table (perfectly matching dataset):

```
CREATE EXTERNAL TABLE student_performance(  
  
    StudentID INT,  
    Age INT,  
    Gender INT,  
    Ethnicity INT,  
    ParentalEducation INT,  
    StudyTimeWeekly DOUBLE,  
    Absences INT,  
    Tutoring INT,  
    ParentalSupport INT,  
    Extracurricular INT,  
    Sports INT,  
    Music INT,  
    Volunteering INT,  
    GPA DOUBLE,  
    GradeClass INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE  
LOCATION '/user/student/student_performance';
```

Submit: Screenshot of table creation.

Task 4 – Explore Dataset Using Hive Queries

- I. Count total students

```
SELECT COUNT(*) FROM student_performance;
```

- II. Compute average GPA

```
SELECT AVG(GPA) AS avg_gpa FROM student_performance;
```

III. Find average study time by parental education level

```
SELECT ParentalEducation,  
       AVG(StudyTimeWeekly) AS avg_study_time  
FROM student_performance  
GROUP BY ParentalEducation;
```

IV. Identify top 5 students with highest GPA

```
SELECT StudentID, GPA  
FROM student_performance  
ORDER BY GPA DESC  
LIMIT 5;
```

Submit: Query outputs + screenshot.

Task 5 – Spark Processing (PySpark)

Using PySpark, create a new derived column called TotalActivities, calculated as:

Sports + Music + Extracurricular + Volunteering

Steps

1. Start PySpark: *pyspark*
2. Load data:

```
data =  
sc.textFile("/user/cloudera/student/student_performance/Student_performanc  
e_data.csv")
```

3. Extract header:

```
header = data.first()
rows = data.filter(lambda row: row != header).map(lambda line: line.split(","))
```

4. Use Row and parse safely

```
from pyspark.sql import Row
def to_row(fields):
    return Row(
        StudentID=int(float(fields[0])),
        Age=int(float(fields[1])),
        Gender=int(float(fields[2])),
        Ethnicity=int(float(fields[3])),
        ParentalEducation=int(float(fields[4])),
        StudyTimeWeekly=float(fields[5]),
        Absences=int(float(fields[6])),
        Tutoring=int(float(fields[7])),
        ParentalSupport=int(float(fields[8])),
        Extracurricular=int(float(fields[9])),
        Sports=int(float(fields[10])),
        Music=int(float(fields[11])),
        Volunteering=int(float(fields[12])),
        GPA=float(fields[13]),
        GradeClass=int(float(fields[14]))
    )
```

```
rows_rdd = rows.map(to_row)
```

5. Create DataFrame: `df = sqlContext.createDataFrame(rows_rdd)`

6. Add TotalActivities column

```
from pyspark.sql.functions import col
df2 = df.withColumn("TotalActivities",
    col("Sports") + col("Music") + col("Extracurricular") + col("Volunteering"))
```

7. Show first 10 rows: `df2.show(10)`

8. Save results:

```
rdd_csv = df2.map(lambda row: ",".join([str(c) for c in row]))
rdd_csv.saveAsTextFile("/user/cloudera/student/student_performance/output_csv/")
```

Submit: Screenshot of `df2.show()` showing *TotalActivities*.

Task 6 – Create Hive Table for Processed Data

Create an external Hive table that reads the Spark output.

```
USE student_db;

CREATE EXTERNAL TABLE student_performance_processed(
  StudentID INT,
  Age INT,
  Gender INT,
  Ethnicity INT,
  ParentalEducation INT,
  StudyTimeWeekly DOUBLE,
  Absences INT,
  Tutoring INT,
  ParentalSupport INT,
  Extracurricular INT,
  Sports INT,
  Music INT,
  Volunteering INT,
  GPA DOUBLE,
  GradeClass INT,
  TotalActivities INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/cloudera/student/student_performance/output/';
```

Verify : *SELECT * FROM student_performance_processed LIMIT 10;*

Submit: Screenshot of the table preview showing TotalActivities.

Task 7 – Visualization (Hue)

Plot average GPA grouped by TotalActivities.

1st verify student db exists in hue, if not create like in task 6, then follow the steps.

1. Run query in Hue:

```
SELECT TotalActivities, AVG(GPA) AS avg_gpa  
FROM student_performance_processed  
GROUP BY TotalActivities  
ORDER BY TotalActivities;
```

2. Choose Bar Chart

- X-axis: TotalActivities
- Y-axis: avg_gpa

Submit: Screenshot of generated chart.

Task 8 – Export Results to Local System

Export processed data from HDFS to the local system.

```
mkdir -p ~/student_processed_output
```

```
hdfs dfs -get /user/cloudera/student/student_performance/output_csv/*  
~/student_processed_output/
```

```
ls ~/student_processed_output/
```

Submit: Screenshot of exported folder.

Task 9 – Final Insights Report

Write a brief analysis (6–8 lines) answering:

- A. Does higher StudyTimeWeekly correlate with higher GPA?
- B. Do students with more TotalActivities tend to have higher GPA?
- C. Which parental education level shows highest average study time?

Submission Instructions

Submit one compiled PDF report containing complete answers to all sub-parts containing.

- Screenshots from Activities
- Answers to every question

File Naming Format:

IS2210_Lab11_IndexNumber
